

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop an Interactive Personal Portfolio Webpage Using HTML, CSS and JavaScript

2. AIM

To develop an interactive personal portfolio webpage using HTML for structure, CSS for styling and JavaScript for dynamic functionality such as a live clock, theme switching and contact-form interaction.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
 - To design a personal portfolio using HTML sections.
 - To apply CSS for layout, colors, spacing, cards and responsive grids.
 - To create navigation links for About, Skills, Projects and Contact sections.
 - To implement a live digital clock using JavaScript.
 - To implement a light/dark theme toggle using JavaScript and CSS classes.
 - To validate and process a contact form using JavaScript.
 - To understand the integration of HTML, CSS and JavaScript in a single webpage.
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4. THEORY

HTML (HyperText Markup Language) provides the structure and content of the webpage. CSS (Cascading Style Sheets) controls presentation, including colors, gradients, spacing, typography, navigation styling, cards and grid layouts. JavaScript adds behavior and interactivity to the webpage.

Important concepts used in this program:

HTML sections: The webpage is divided into navigation, hero, About, Skills, Projects, Contact and Footer sections.

CSS Grid: The Skills and Projects areas use responsive grid layouts with auto-fit and minimum column widths.

Fixed navigation: The navigation bar remains at the top of the page using position: fixed.

Theme switching: JavaScript toggles the dark class on the body to change the page appearance.

Live clock: JavaScript creates a Date object and updates the displayed time every second.

Form handling: The submit event is prevented from performing a normal page reload; a thankyou message is displayed and the form is reset.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
 2. Add the HTML5 document declaration and metadata.
 3. Set the webpage title as Almas Portfolio.
 4. Create a fixed navigation bar with links to the portfolio sections.
 5. Create the hero section with the student's name and professional interests.
 6. Add a live clock container and a Toggle Theme button.
 7. Create About, Skills, Projects and Contact sections.
 8. Use CSS cards and responsive grids to arrange skills and projects.
 9. Create a contact form with name, email and message fields.
 10. Create JavaScript functions for theme switching and form submission.
 11. Create the updateClock function and call it every second using setInterval.
 12. Add a footer containing the copyright information.
 13. Save the file with the .html extension.
 14. Open the file in a web browser and test all navigation and interactive features.
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6. PROGRAM

static_webpage.html



```
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>Almas
Portfolio</title>

<style>

*{  margin:0;  padding:0;
box-sizing:border-box;  font-
family:'Segoe UI',sans-serif;
scroll-behavior:smooth;
}

body{  background:linear-
gradient(135deg,#0f172a,#1e293b);  color:white;
}

/* Navbar */

nav{
position:fixed;
top:0;
width:100%;
display:flex;
```



```
nav ul{
display:flex; list-
style:none;
}

nav ul li{ margin-
left:25px;
}

nav ul li a{
color:white; text-
decoration:none;
transition:.3s;
}

nav ul li a:hover{
color:#38bdf8;
}

/* Hero Section */

.hero{ height:100vh;
display:flex; justify-
content:center; align-
items:center; text-
align:center; flex-
direction:column;
}
```



```
margin-top:15px;  
font-size:1.2rem;  
}  
  
.btn{  margin-top:25px;  
padding:12px 25px;  
background:#38bdf8;  
color:black;  
border:none;  border-  
radius:30px;  
cursor:pointer;  font-  
weight:bold;  
transition:.3s;  
}
```



```
.btn:hover{
transform:scale(1.05);
}

/* Sections */

section{  padding:100px
10%;
}

.section-title{
```

```
border-radius:20px; line-
height:1.8;
}

/* Skills */

.skills{ display:grid; grid-template-columns:repeat(auto-
fit,minmax(200px,1fr)); gap:20px;
}

.skill{
background:rgba(255,255,255,0.08);
padding:25px; border-radius:15px;
text-align:center; transition:.3s;
}

.skill:hover{ transform:translateY(-
10px);
}

/* Projects */

.projects{ display:grid; grid-template-
columns:repeat(auto-fit,minmax(280px,1fr));
```



```
.project:hover{ transform:translateY(-10px);
}
```

```
.project h3{
color:#38bdf8; margin-bottom:10px;
}
```

```
/* Contact */
```

```
form{ max-width:600px;
margin:auto;
}
```

```
input,textarea{
width:100%;
padding:15px;
margin:10px 0;
border:none; border-radius:10px;
}
```



```
textarea{
height:150px;
}
```

A large solid dark blue rectangular area in the center of the page, containing the CSS code snippet.





```
font-size:20px;
}

.dark{
background:#ffffff;
color:#111;
}

.dark nav{
background:rgba(0,0,0,0.08);
}

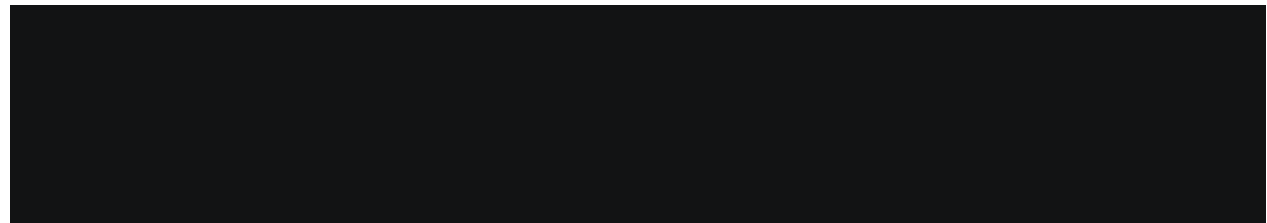
.dark .about-card,
.dark .skill, .dark .project{
background:rgba(0,0,0,0.08);
}

</style>
</head>

<body>

<nav>
  <h2>ALMAS</h2>

  <ul>
```

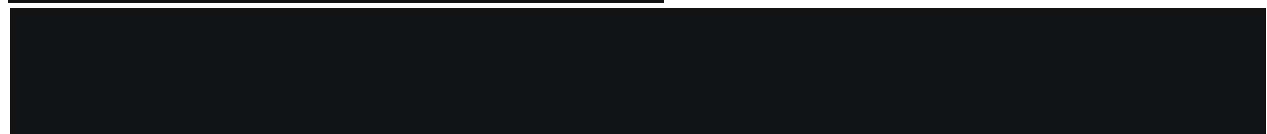


```
<div id="clock"></div>

<button class="btn" onclick="toggleTheme()">
  Toggle Theme
</button>
</section>

<section id="about">
  <h2 class="section-title">About Me</h2>

  <div class="about-card">
    I am a passionate student interested in
    Web Development, Machine Learning and
    Software Engineering. I enjoy building dynamic
    applications using HTML, CSS, JavaScript and
    Python.
  </div>
</section>
```



```
<section id="skills">
  <h2 class="section-title">Skills</h2>

  <div class="skills">
    <div class="skill">HTML5</div>
    <div class="skill">CSS3</div>
    <div class="skill">JavaScript</div>
    <div class="skill">Python</div>
```

```
<h3>Iris Prediction</h3>
<p>KNN based flower prediction system.</p>
</div>

<div class="project">
  <h3>House Price Prediction</h3>
  <p>Regression model for predicting house prices.</p>
</div>

<div class="project">
  <h3>Student Management</h3>
  <p>Dynamic CRUD application using JavaScript.</p>
</div>

</div>
</section>

<section id="contact">
  <h2 class="section-title">Contact Me</h2>

  <form onsubmit="sendMessage(event)">
    <input type="text" id="name" placeholder="Your Name" required>

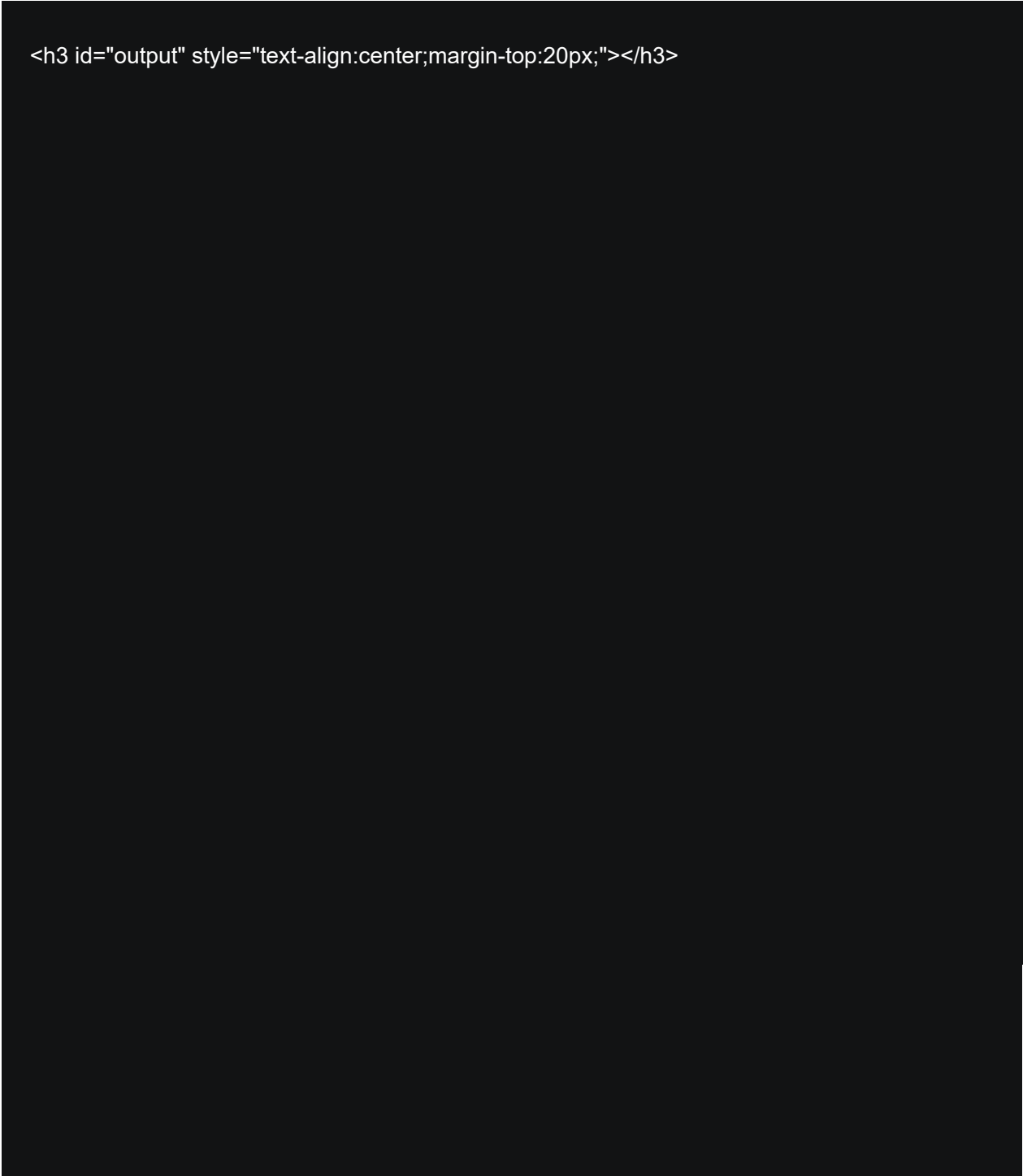
    <input type="email" id="email" placeholder="Your Email" required>

    <textarea id="msg" placeholder="Your Message"></textarea>

    <button class="btn">Send Message</button>
  </form>
```



`<h3 id="output" style="text-align:center;margin-top:20px;"></h3>`





```
function toggleTheme(){
document.body.classList.toggle("dark");
}

function sendMessage(event){
event.preventDefault();

    let name =
document.getElementById("name").value;

    document.getElementById("output").innerHTML =
    "☒ Thank you " + name + "! Message sent successfully.";

    document.querySelector("form").reset();
}

function updateClock(){
let now = new Date();

    document.getElementById("clock").innerHTML =
now.toLocaleTimeString();
}

setInterval(updateClock,1000);

</script>

</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
<title>	Specifies the title of the webpage.
<body>	Contains the visible webpage content.
<header>	Creates the header section of the webpage.
<nav>	Contains navigation links.
.hero	Creates the full-screen introductory section
.skills and .projects	Create responsive CSS Grid layouts for portfolio cards
.btn	Styles the interactive buttons and adds a hover animation.
toggleTheme()	Adds or removes the dark class from the body element.
sendMessage(event	Prevents the default form submission, displays a personalized confirmation and resets the form.

updateClock()	Reads the current time and displays it in the clock element.
setInterval(updateClock,1000)	Runs the clock update once every second.
document.querySelector('form').reset()	Clears the form after the message is processed.

8. TEST CASES AND EXECUTION DATA

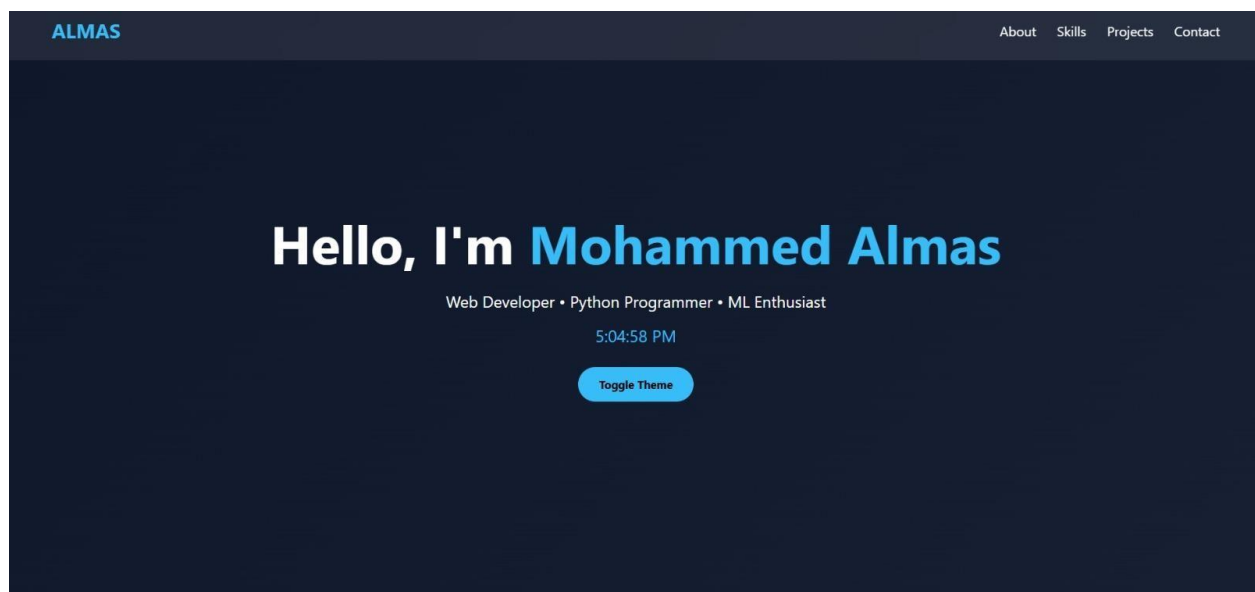
I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Click About/Skills/Projects/Contact	Corresponding section should be displayed	Section displayed	Pass
TC-02	Click Toggle Theme	Page theme should change	Theme changed	Pass
TC-03	Observe live clock	Current time should update every second	Clock updated	Pass
TC-04	Enter name, email and message; submit form	Thank-you message should be displayed	Message displayed	Pass

TC-05	Submit with empty Name	Browser should require the Name field	Required validation shown	Pass
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9. OUTPUT

OUTPUT 1 : Navbar + Hero



Hello, I'm Mohammed Almas

Web Developer • Python Programmer • ML Enthusiast

5:05:08 PM

Toggle Theme

OUTPUT 2 : About Me

About Me

I am a passionate student interested in Web Development, Machine Learning and Software Engineering. I enjoy building dynamic applications using HTML, CSS, JavaScript and Python.

OUTPUT 3: Projects

Projects

Iris Prediction

KNN based flower prediction system.

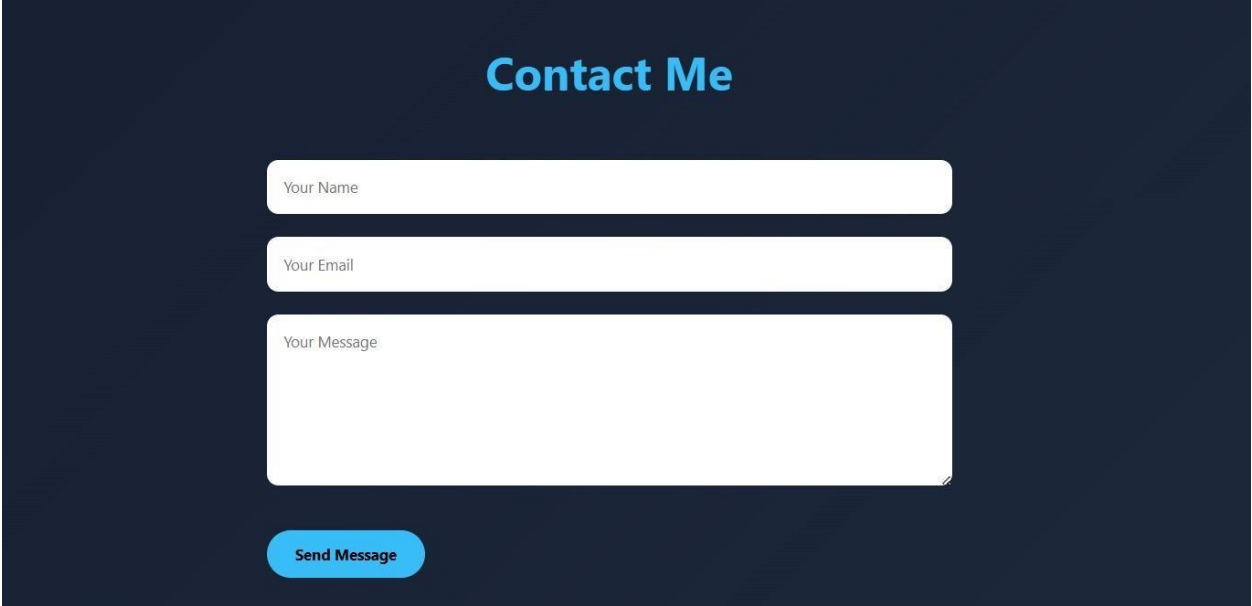
House Price Prediction

Regression model for predicting house prices.

Student Management

Dynamic CRUD application using JavaScript.

OUTPUT 4: Contact Me

A contact form titled "Contact Me" in a light blue font. It features three white input fields with rounded corners: "Your Name", "Your Email", and "Your Message". Below the message field is a light blue "Send Message" button with rounded corners. The entire form is centered on a dark navy blue background.

Contact Me

Your Name

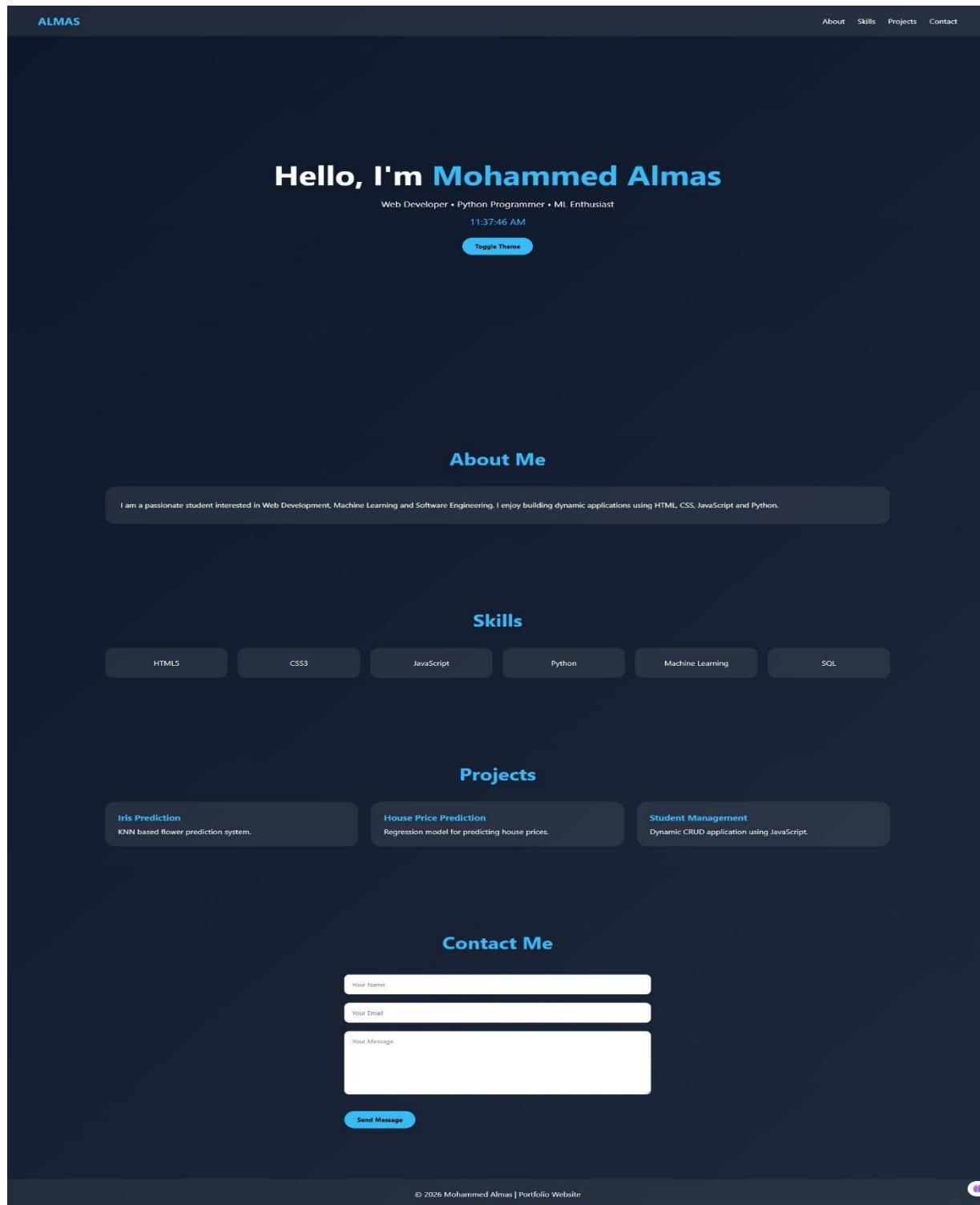
Your Email

Your Message

Send Message

OUTPUT 5: Footer

Complete Webpage Output:



10. RESULT

Thus, an interactive personal portfolio webpage was successfully developed using HTML, CSS and JavaScript. The webpage includes navigation, a hero section, About, Skills, Projects and

Contact sections. The live clock, theme toggle and contact-form interaction were implemented and tested successfully.
